Black-Scholes Model: Numerical Study

Finite Differences, Monte Carlo, and Delta Hedging

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PRP Report

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Problem Statement: The Black-Scholes PDE

We solve the parabolic PDE for the option price $V(S,t)$:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Boundary conditions (European Call):

- Terminal payoff: $V(S,T) = \max(S-K, 0)$
- Left boundary: $V(0,t) = 0$
- Right boundary: $V(S_{\max}, t) \approx S_{\max} - Ke^{-r(T-t)}$

Notation:

S	Asset price (spot)	r	Risk-free interest rate
t	Current time	σ	Volatility (diffusion coefficient)
T	Maturity (expiration)	$V(S,t)$	Option price (unknown function)
K	Strike price	Δ	Delta (hedge ratio) $= \partial V/\partial S$

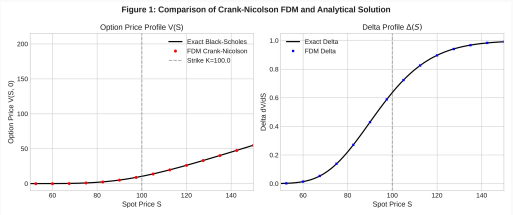
Analytical Benchmark (Ground Truth)

The closed-form solution for a European Call:

$$C(S,t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$$

where $\Phi(\cdot)$ is the standard normal CDF, and

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$



Method 1: Crank-Nicolson FDM

Discretization: Grid $S_i = i\Delta S$, $t_j = j\Delta t$. Let $V_i^j = V(S_i, t_j)$.

Crank-Nicolson Scheme (average of implicit and explicit):

$$\frac{V_i^{j+1} - V_i^j}{\Delta t} + \frac{1}{2}\left[\mathcal{L}V_i^{j+1} + \mathcal{L}V_i^j\right] = 0$$

Matrix form: $(I + \frac{\Delta t}{2}L)V^{j+1} = (I - \frac{\Delta t}{2}L)V^j$.

Properties:

- Order: $O(\Delta t^2 + \Delta S^2)$
- Unconditionally stable (A-stable)
- Tridiagonal system $\rightarrow$ solved in $O(M)$ via Thomas algorithm

Crank-Nicolson: Matrix coefficients

Coefficients (from central differences, $S_i = i\Delta S$):

$$a_i = \frac{\Delta t}{4}(\sigma^2 i^2 - ri), \quad b_i = -\frac{\Delta t}{2}(\sigma^2 i^2 + r), \quad c_i = \frac{\Delta t}{4}(\sigma^2 i^2 + ri)$$

System:

$$-a_i V_{i-1}^{j+1} + (1 - b_i) V_i^{j+1} - c_i V_{i+1}^{j+1} = a_i V_{i-1}^{j} + (1 + b_i) V_i^{j} + c_i V_{i+1}^{j}$$

FDM Results: Convergence

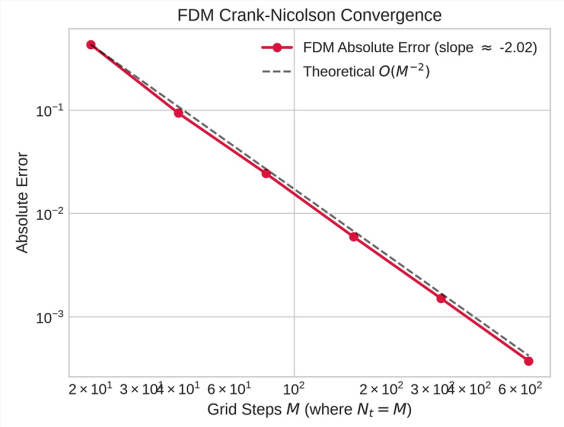


Figure 1: FDM error decay: empirical slope ≈ -2.02 (theoretical $O(M^{-2})$).

The empirical slope perfectly matches the theoretical second-order convergence.

Method 2: Monte Carlo

GBM Simulation (risk-neutral measure):

$$S_T^{(k)} = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right) T + \sigma\sqrt{T} Z^{(k)}\right], \quad Z^{(k)} \sim \mathcal{N}(0,1)$$

Price Estimator:

$$\widehat{C} = e^{-rT} \frac{1}{N} \sum_{k=1}^{N} \max(S_T^{(k)} - K, 0)$$

Properties:

- Convergence: $O(N^{-1/2})$ from the Central Limit Theorem.
- Antithetic Variates: Pair $(Z, -Z)$ to reduce variance.
- Empirical variance reduction: $\approx 2.03\times$.

Monte Carlo Results: Convergence

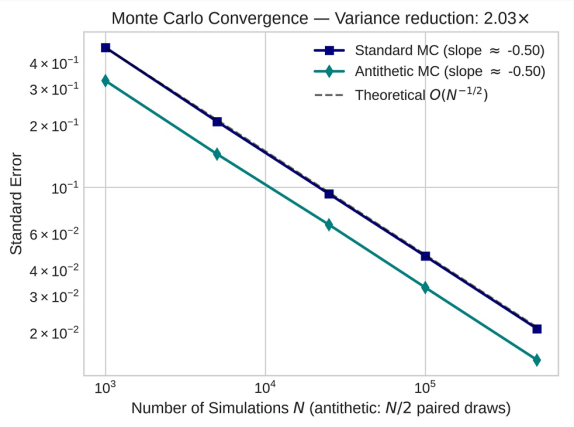


Figure 2: MC error decay: empirical slope ≈ -0.50 (theoretical $O(N^{-1/2})$).

Both standard and antithetic MC follow the $1/\sqrt{N}$ law. Antithetic reduces variance by a factor of 2.

Delta Hedging Mechanics

Goal: Eliminate the directional risk of a sold option.

Idea: Hold Δ shares of the underlying asset. If S changes by dS, the option changes by ΔdS, and the shares change by ΔdS (net change ≈ 0).

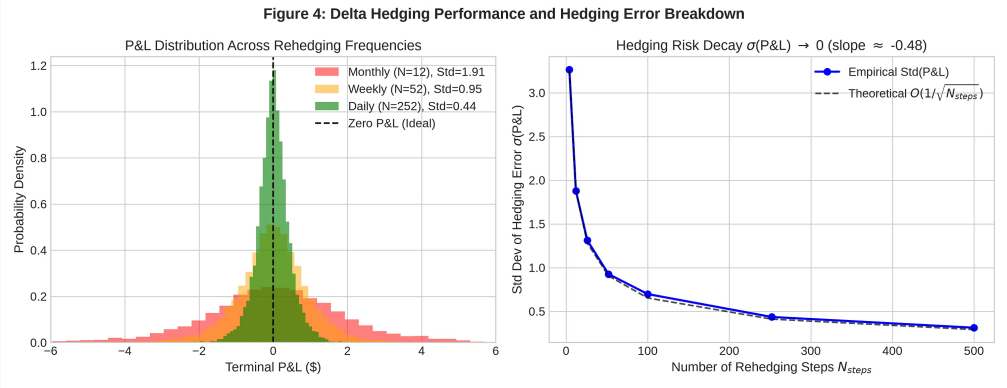
Algorithm (Discrete Rebalancing):

1. **At $t = 0$:** Sell 1 Call for C_0. Compute $\Delta_0 = \partial V/\partial S$. Buy Δ_0 shares. Bank cash: $B_0 = C_0 - \Delta_0 S_0$.
2. **At each step t_j:** Recompute Δ_j. Buy/sell $(\Delta_j - \Delta_{j-1})$ shares. Pay interest on the bank account $B_j = B_{j-1} e^{r\Delta t}$.
3. **At maturity T:** Liquidate shares. Pay the option payoff. Compute the terminal P&L.

Terminal P&L:

$$P\&L = B_T + \Delta_T S_T - \max(S_T - K, 0)$$

Delta Hedging: Error Decay



Error decay: $\sigma(P\&L) \propto 1/\sqrt{N_{\text{steps}}}$ (empirical slope ≈ -0.48).

Delta Hedging: Error Decay

Theoretical error bound:

$$P\&L \approx -\frac{1}{2}\sum_j \Gamma_j S_j^2 \sigma^2 \left[(\Delta W_j)^2 - \Delta t\right]$$

where $\Gamma = \partial^2 V/\partial S^2$.
This gives $\mathrm{Var}(P\&L) \propto \frac{1}{N_{\text{steps}}}$, hence $\sigma(P\&L) \propto \frac{1}{\sqrt{N_{\text{steps}}}}$.

Hedging Trade-off: FDM vs MC Delta

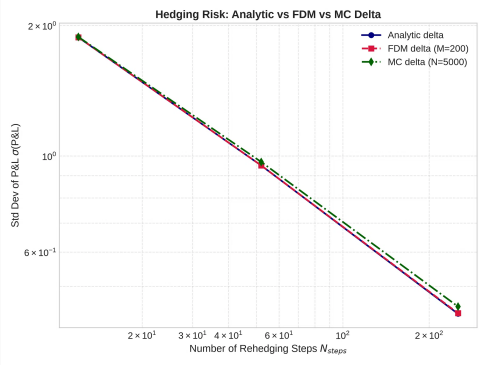


Figure 3: FDM delta matches analytics perfectly.

FDM is deterministic and smooth; MC is universal but introduces stochastic noise.

Validation Summary (Base Case)

$S_0 = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.2$. Exact price: 10.4506.

Method	Price	Absolute Error	Order
Analytical	10.4506	—	—
FDM ($M = 200$)	10.4544	3.85×10^{-3}	$O(M^{-2})$
MC ($N = 10^5$)	10.4739	2.3×10^{-2}	$O(N^{-1/2})$
MC + Antithetic ($N = 10^5$)	10.4529	2.3×10^{-3}	$O(N^{-1/2})$

Hedging error decay (empirical slope ≈ -0.48):

Monthly ($N = 12$)	Weekly ($N = 52$)	Daily ($N = 252$)
$\sigma(P\&L) = 1.91$	$\sigma(P\&L) = 0.95$	$\sigma(P\&L) = 0.44$

Code Architecture

Structure:

- `bs_engine.py` — Analytics + FDM
- `bs_monte_carlo.py` — MC + Hedging
- `fdm_delta.py` / `mc_delta.py` — Adapters
- `benchmarks.py` — Figures generation

Code

`https://github.com/batarikiri/black-scholes-numerical-engine`

Done:

- FDM and MC validated against analytics.
- FDM delta replaces analytical delta without loss.
- MC delta introduces noise (trade-off).

Other tasks:

- American options (LCP)
- Stochastic Heston Volatility (2D PDE)